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Scrapy the Agent!

Going from a disease of interest to extracted risk metrics without human intervention.

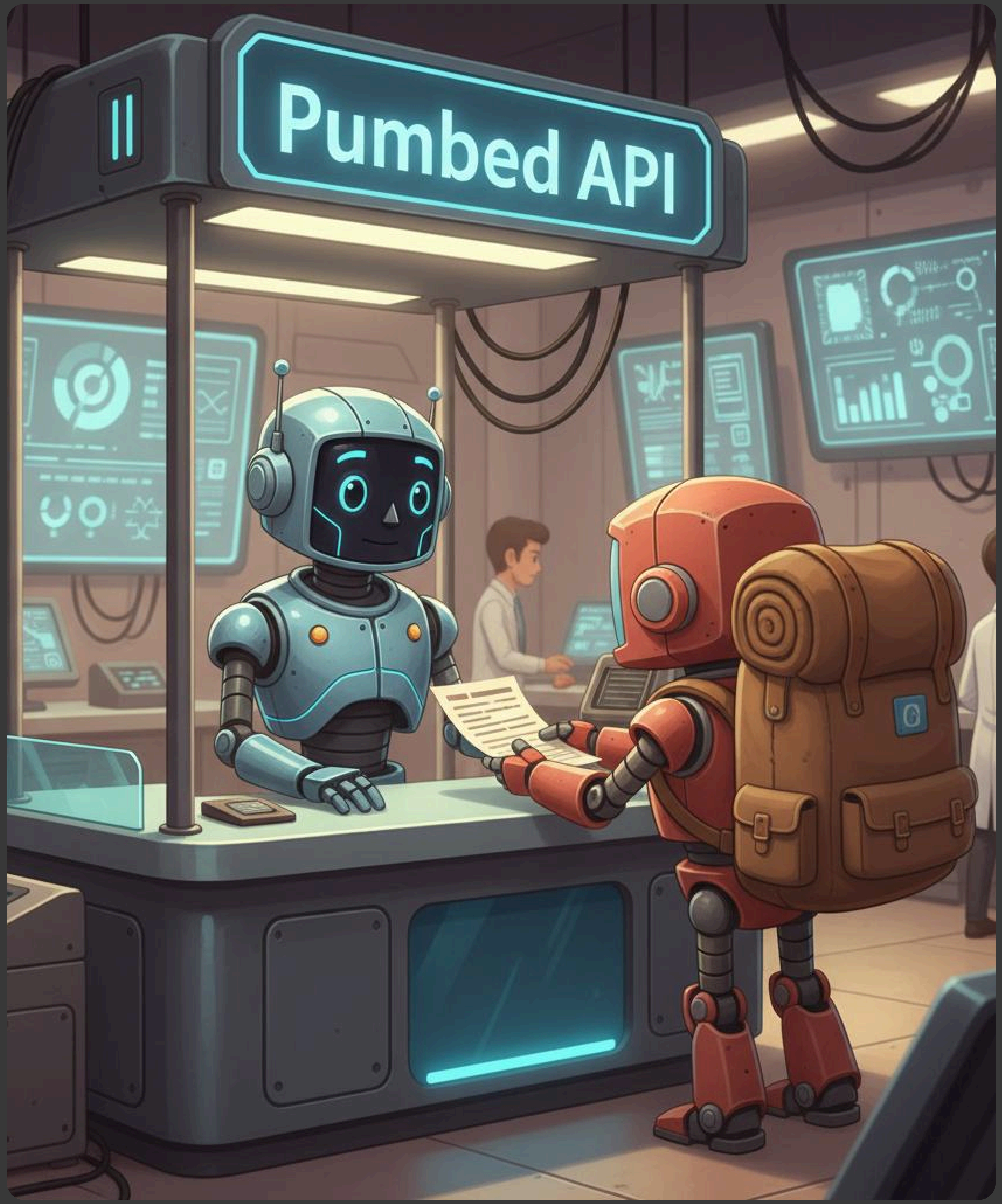
Step 0:
Starting the yourney

```
result = agent.invoke(
                {
                    "disease_of_interest": "CVD",
                    "metrics_count": 0,
                    "max_metrics": 100,
                    "max_papers": 300,
                    "max_queries": 30,
                    "checked_dois": [],
                    "tried_queries": []
                },
                {"recursion_limit": 2000}
            )
```

Step 1:

PubMed search

- Autonomous query creation based on disease of interest
- Queries get refined iteratively to uncover all relevant papers



Step 1:

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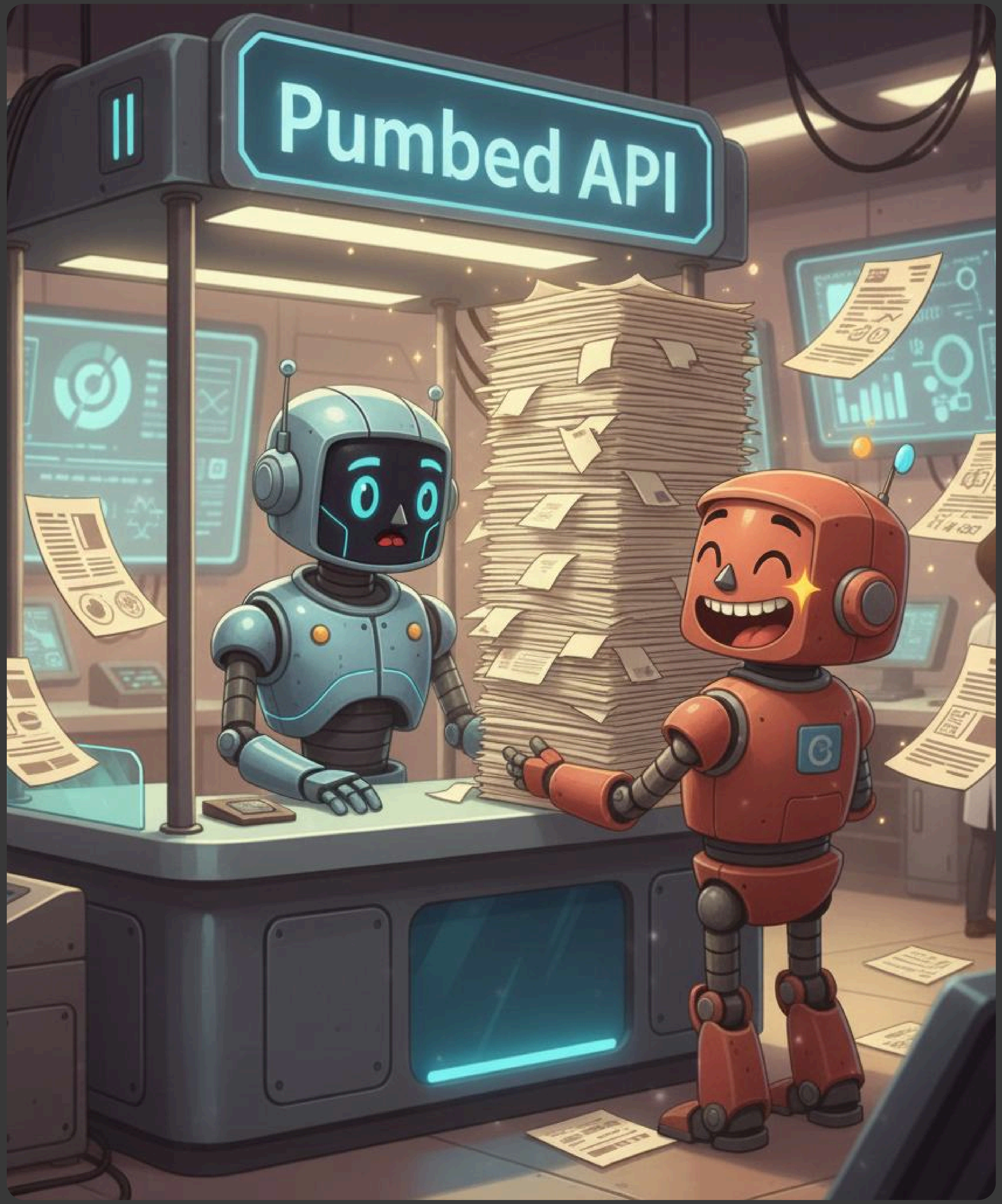


Step 2:

Relevance checking

- Only keep papers that are relevant to the disease of interest
- Checking is done based on the abstract

Step 3:

Automated PDF download

- Open access papers get downloaded automatically

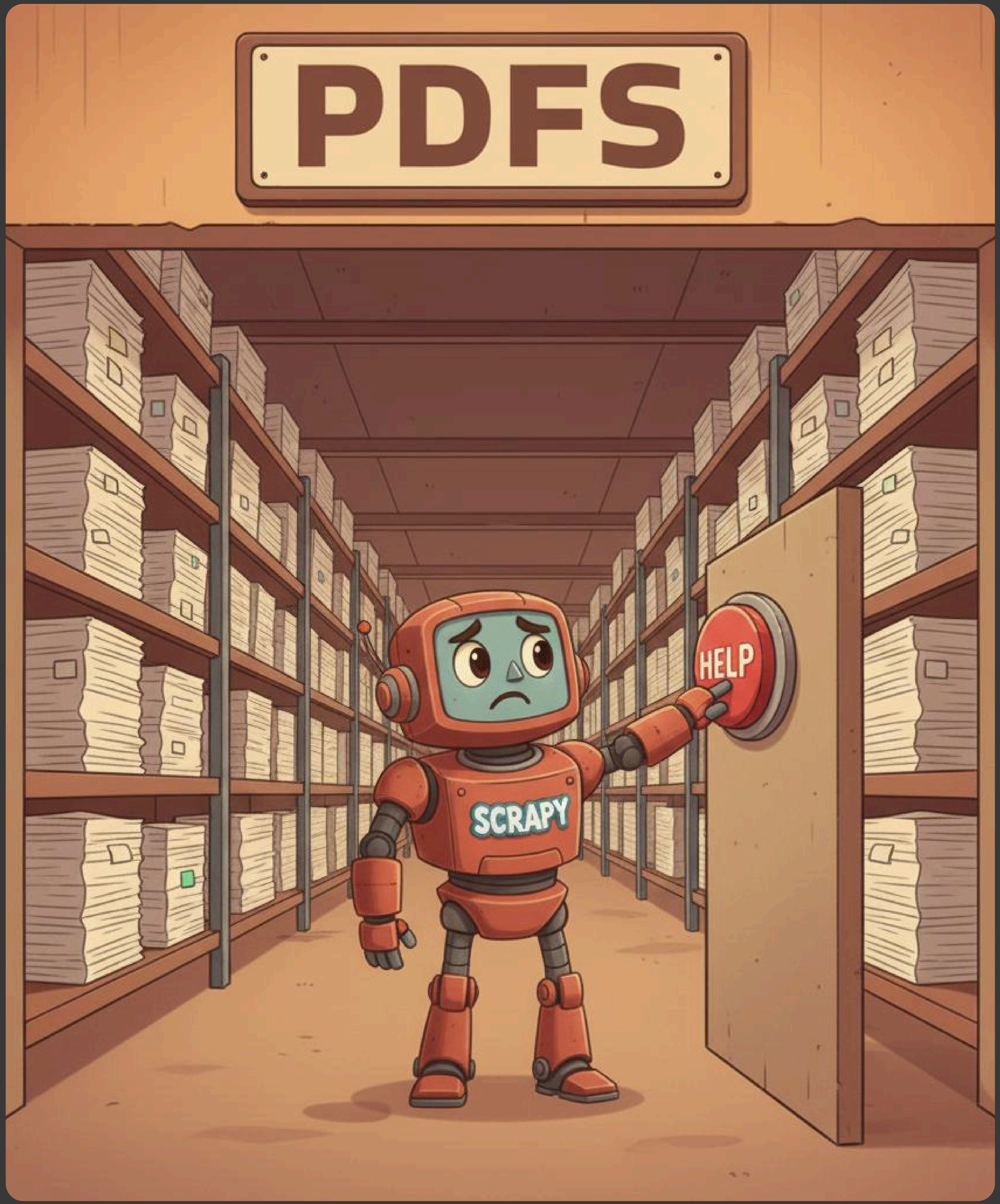
```
pdf_from_doi = PDFFromDOI()
path = pdf_from_doi.download("10.1016/S2468-2667(19)30155-0")
path = pdf_from_doi.download("10.1001/jamacardio.2016.2415")
path = pdf_from_doi.download("10.1155/2021/6636856")
    print(path)
```

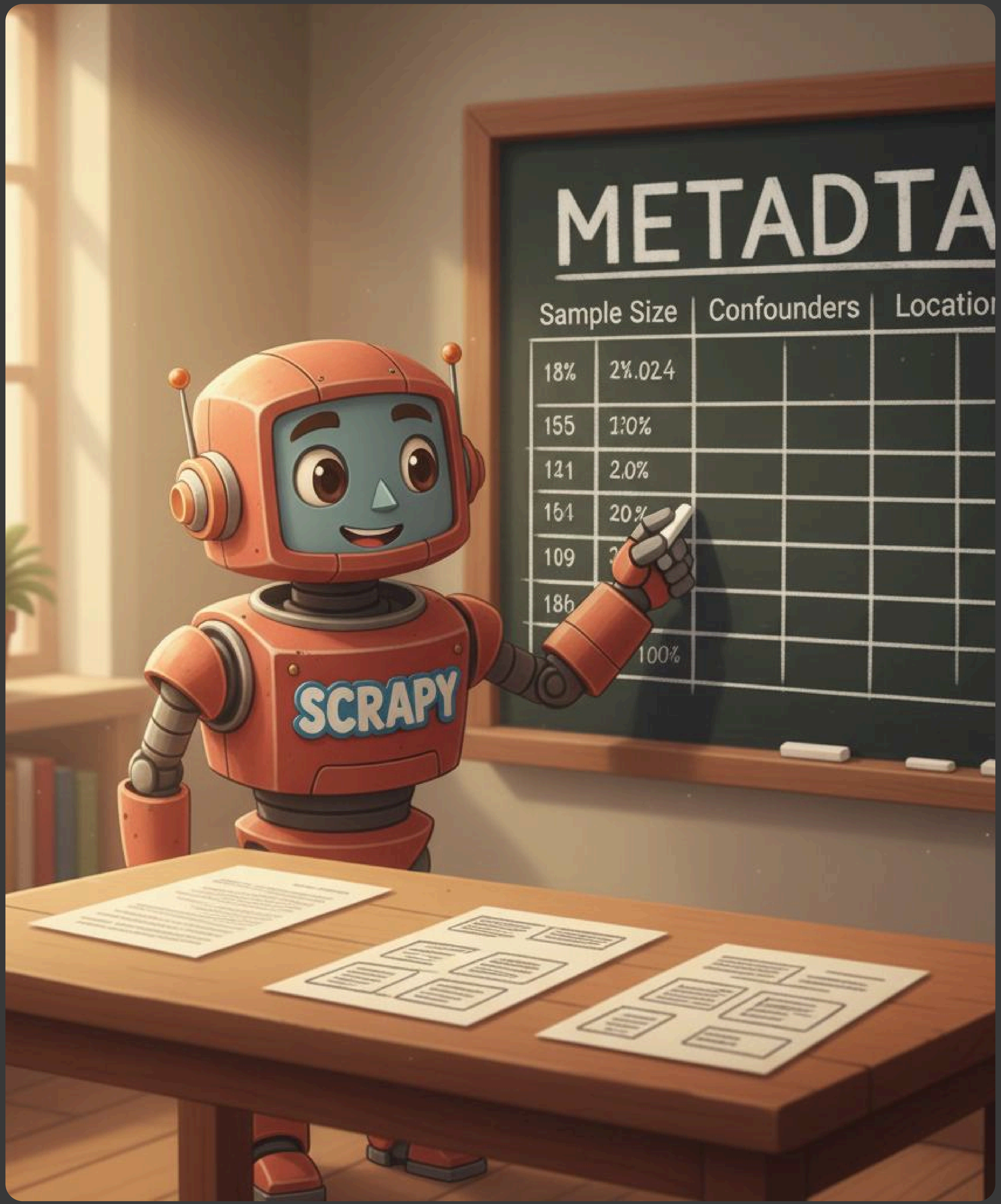
Step 3:

Automated PDF download

- Open access papers get downloaded automatically
- Closed source papers get stored in a CSV for manual downloads

```
pdf_from_doi = PDFFromDOI()
path = pdf_from_doi.download("10.1016/S2468-2667(19)30155-0")
path = pdf_from_doi.download("10.1001/jamacardio.2016.2415")
path = pdf_from_doi.download("10.1155/2021/6636856")
    print(path)
```





Step 4:

Metadata extraction

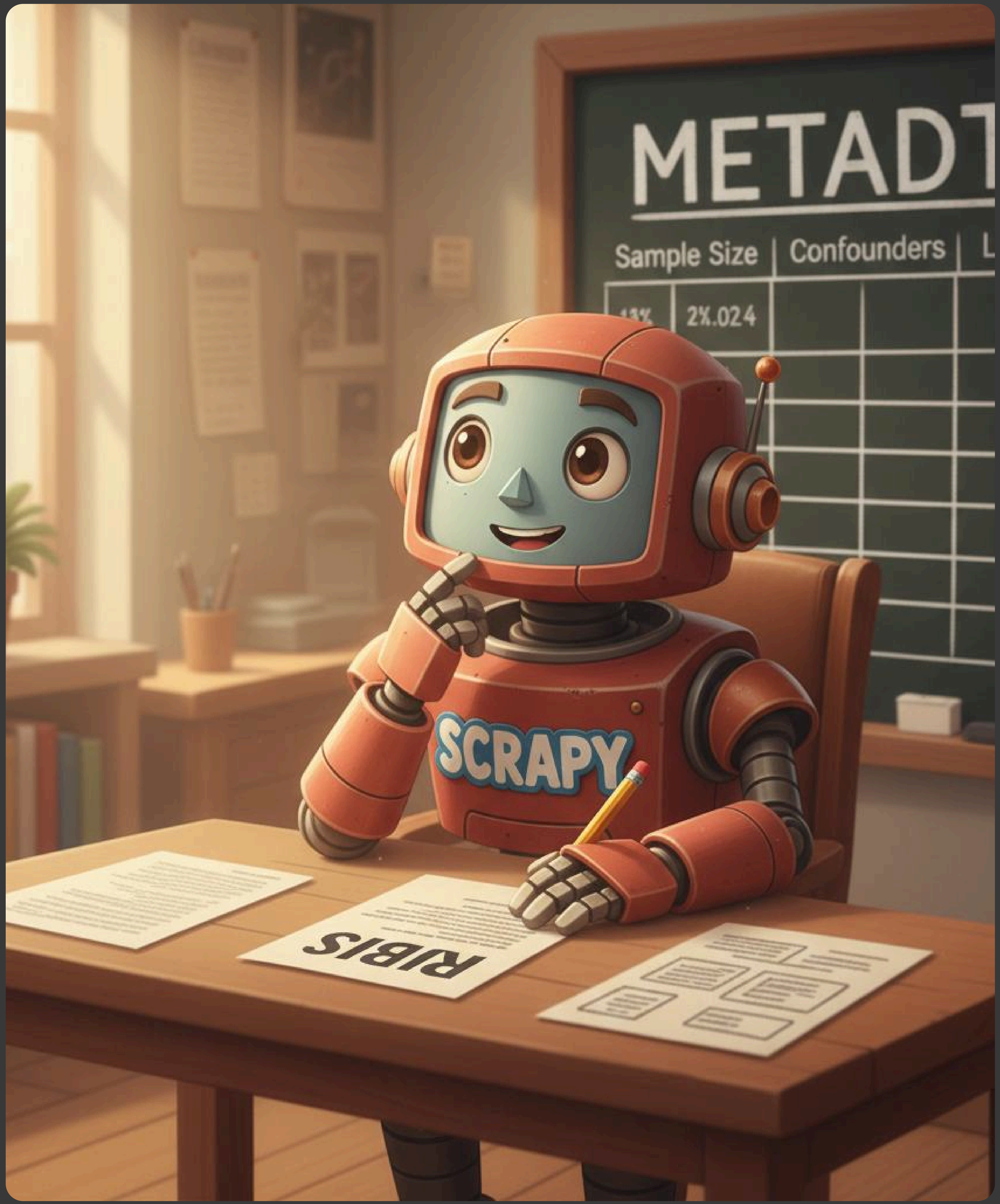
```
meta_analysis_metadata = [
    {
        "n_of_included_studies": "12",
        "sample_size": "324567",
        "geography": "USA, UK, Europe",
        "confounder_vars": "Age, BMI, smoking status, socioeconomic status, hormone replacement therapy"
    },
]
```

Step 5:

ROBIS (Whiting et al., 2016) analysis

Judging the quality of each scientific paper individually

```
system_prompt = """
You are an AI agent tasked with evaluating the risk of
bias in a meta-analysis paper using the ROBIS (Risk Of
Bias In Systematic reviews) tool..."""
```



Step 6:

Risk-data extraction

```
metrics = [
    {
        "menopause_timing_definition": "40-50 vs >=51",
        "health_outcome": "All cause dementia",
        "metric_type": "HR",
        "metric_value": "1.45",
        "ci_95": "1.20-1.75"
    }
]
```

Scrapies Statistics:

Worked for

5 hours 22 min

Checked

1655 papers

Extracted

584 risk metrics

69	highest vs lowest category	All cause dementia	ES	0.77	0.62-0.95	https://doi.org/10.1016/j.psyneuen.2016.08.003	2016-Nov	13	19449	Greece, Can	Age, educatio	['Georgakis, I	Low	9
70	highest vs lowest category	Alzheimer's disease	ES	1.23	0.58-2.62	https://doi.org/10.1016/j.psyneuen.2016.08.003	2016-Nov	13	19449	Greece, Can	Age, educatio	['Georgakis, I	Low	9
71	N/A (study focused on vaginal estrogen therapy use, not me	Breast cancer recurrence (general popu	RR	0.87	0.67-1.11	https://doi.org/10.61622/rbgo/2025rbgo46	2025	7	118659	USA, Australi	N/A	['Santos, Gab	Low	10
72	N/A (study focused on vaginal estrogen therapy use, not me	Breast cancer recurrence (with aromat	RR	2.59	0.74-9.09	https://doi.org/10.61622/rbgo/2025rbgo46	2025	7	118659	USA, Australi	N/A	['Santos, Gab	Low	10
73	N/A (study focused on vaginal estrogen therapy use, not me	All-cause mortality (general population)	RR	0.8	0.75-0.86	https://doi.org/10.61622/rbgo/2025rbgo46	2025	7	118659	USA, Australi	N/A	['Santos, Gab	Low	10
74	N/A (study focused on vaginal estrogen therapy use, not me	All-cause mortality (with aromatase inh	RR	0.87	0.69-1.10	https://doi.org/10.61622/rbgo/2025rbgo46	2025	7	118659	USA, Australi	N/A	['Santos, Gab	Low	10
75	<40 vs 49-54 (normal ANM)	All cause mortality	RR	1.39	1.10-1.77	https://doi.org/10.3109/13697137.2015.1094784	2016	7	397602	USA, China, N	Age, educatio	['Tao, X-Y', 'Zu	Low	8
76	<40 vs 49-54 (normal ANM)	Ischemic heart disease (IHD) mortality	RR	1.48	1.02-2.16	https://doi.org/10.3109/13697137.2015.1094784	2016	7	397602	USA, China, N	Age, educatio	['Tao, X-Y', 'Zu	Low	8
77	40-44 vs 49-54 (normal ANM)	Ischemic heart disease (IHD) mortality	RR	1.09	1.00-1.18	https://doi.org/10.3109/13697137.2015.1094784	2016	7	397602	USA, China, N	Age, educatio	['Tao, X-Y', 'Zu	Low	8
78	<40 vs 49-54 (normal ANM)	Cardiovascular disease (CVD) mortality	RR	1.24	0.98-1.58	https://doi.org/10.3109/13697137.2015.1094784	2016	7	397602	USA, China, N	Age, educatio	['Tao, X-Y', 'Zu	Low	8
79	<40 vs 50-52 (approximated from normal range)	All cause mortality	RR	1.18	1.07-1.29	https://doi.org/10.1016/j.ijcard.2015.10.092	2016-Jan-15	10	149373	USA, Norway	Age at baseli	['Gong, Dand	Low	9
80	<40 vs 50-52 (approximated from normal range)	CHD mortality	RR	1.11	1.03-1.20	https://doi.org/10.1016/j.ijcard.2015.10.092	2016-Jan-15	10	149373	USA, Norway	Age at baseli	['Gong, Dand	Low	9
81	<40 vs 50-52 (approximated from normal range)	Cardiovascular mortality	RR	1.06	0.92-1.21	https://doi.org/10.1016/j.ijcard.2015.10.092	2016-Jan-15	10	149373	USA, Norway	Age at baseli	['Gong, Dand	Low	9
82	<40 vs 50-52 (approximated from normal range)	Stroke mortality	RR	0.94	0.86-1.04	https://doi.org/10.1016/j.ijcard.2015.10.092	2016-Jan-15	10	149373	USA, Norway	Age at baseli	['Gong, Dand	Low	9
83	<40 vs 50-52	All cause mortality	RR	1.24	1.10-1.39	https://doi.org/10.1016/j.ijcard.2015.10.092	2016-Jan-15	10	149373	USA, Norway	Age at baseli	['Gong, Dand	Low	9
84	<40 vs >=40	All cause mortality	HR	1.53	1.13-2.08	https://doi.org/10.1089/jwh.2023.0189	2023-Sep	N/A	1210	USA	Age, race, ma	['Xing, Zailing	Low	9
85	per 1-year increase in age at menopause	All cause mortality	HR	0.96	0.94-0.98	https://doi.org/10.1089/jwh.2023.0189	2023-Sep	N/A	1210	USA	Age, race, ma	['Xing, Zailing	Low	9
86	per 1-year increase in reproductive life span	All cause mortality	HR	0.96	0.94-0.98	https://doi.org/10.1089/jwh.2023.0189	2023-Sep	N/A	1210	USA	Age, race, ma	['Xing, Zailing	Low	9
87	<40 vs >=51	All cause mortality (adjusted)	HR	1.08	1.03-1.14	https://doi.org/10.1155/2021/6636856	2021	16	321233	USA, Netherl	Age, hormone	['Huan, Luyac	Low	10
88	<40 vs >=51	All cause mortality (adjusted)	RR	1.05	1.01-1.08	https://doi.org/10.1155/2021/6636856	2021	16	321233	USA, Netherl	Age, hormone	['Huan, Luyac	Low	10
89	<40 vs >=51	All cause mortality (unadjusted)	HR	1.12	1.05-1.19	https://doi.org/10.1155/2021/6636856	2021	16	321233	USA, Netherl	Age, hormone	['Huan, Luyac	Low	10
90	<40 vs >=51	All cause mortality (unadjusted)	RR	1.03	1.00-1.06	https://doi.org/10.1155/2021/6636856	2021	16	321233	USA, Netherl	Age, hormone	['Huan, Luyac	Low	10
91	<40 vs >=51	Cardiovascular mortality (adjusted)	HR	1.01	0.95-1.09	https://doi.org/10.1155/2021/6636856	2021	16	321233	USA, Netherl	Age, hormone	['Huan, Luyac	Low	10
92	<40 vs >=51	Cardiovascular mortality (adjusted)	RR	1.08	0.77-1.51	https://doi.org/10.1155/2021/6636856	2021	16	321233	USA, Netherl	Age, hormone	['Huan, Luyac	Low	10
93	<40 vs >=51	Cardiovascular mortality (unadjusted)	HR	1.04	1.00-1.13	https://doi.org/10.1155/2021/6636856	2021	16	321233	USA, Netherl	Age, hormone	['Huan, Luyac	Low	10
94	40-44 vs 50-54	All cause mortality (adjusted)	HR	1.12	0.96-1.31	https://doi.org/10.1155/2021/6636856	2021	16	321233	USA, Netherl	Age, hormone	['Huan, Luyac	Low	10
95	40-44 vs 50-54	All cause mortality (adjusted)	RR	1.03	0.96-1.10	https://doi.org/10.1155/2021/6636856	2021	16	321233	USA, Netherl	Age, hormone	['Huan, Luyac	Low	10
96	45-49 vs 50-54	All cause mortality (adjusted)	HR	1.05	1.00-1.10	https://doi.org/10.1155/2021/6636856	2021	16	321233	USA, Netherl	Age, hormone	['Huan, Luyac	Low	10
97	45-49 vs 50-54	All cause mortality (adjusted)	RR	1.02	1.01-1.04	https://doi.org/10.1155/2021/6636856	2021	16	321233	USA, Netherl	Age, hormone	['Huan, Luyac	Low	10
98	<40 vs >=51	All cause mortality (adjusted)	HR	1.1	1.01-1.21	https://doi.org/10.1155/2021/6636856	2021	16	321233	USA, Netherl	Age, hormone	['Huan, Luyac	Low	10
99	<40 vs >=51	All cause mortality (adjusted)	RR	1.34	1.08-1.66	https://doi.org/10.1155/2021/6636856	2021	16	321233	USA, Netherl	Age, hormone	['Huan, Luyac	Low	10
100	<40 vs >=51	Cardiovascular mortality (adjusted)	HR	1.09	1.00-1.19	https://doi.org/10.1155/2021/6636856	2021	16	321233	USA, Netherl	Age, hormone	['Huan, Luyac	Low	10
101	<40 vs >=50	Premature mortality (adjusted)	HR	1.6	1.28-2.00	https://doi.org/10.3390/metabo14110571	2024-Oct-24	N/A	33687	UK	Age, race, ed	['Yang, Zeping	Low	9
102	40-49 vs >=50	Premature mortality (adjusted)	HR	1.17	1.04-1.30	https://doi.org/10.3390/metabo14110571	2024-Oct-24	N/A	33687	UK	Age, race, ed	['Yang, Zeping	Low	9
103	per 5.13-year increase in age at menopause	Premature mortality (adjusted)	HR	0.89	0.84-0.93	https://doi.org/10.3390/metabo14110571	2024-Oct-24	N/A	33687	UK	Age, race, ed	['Yang, Zeping	Low	9
104	per SD increment in metabolomic signature	Premature mortality (adjusted)	HR	0.79	0.75-0.83	https://doi.org/10.3390/metabo14110571	2024-Oct-24	N/A	33687	UK	Age, race, ed	['Yang, Zeping	Low	9

The Final Result!

https://github.com/EliasSchlie/quantifying-the-value-of-delayed-ovarian-aging (MIT licence)

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Health Methodology

01

Choose risk values

Manually go through the risk metrics for each disease. Pick based on cohort constraints and study quality. Manually check each value.

02

Find annual incidences per 100,000

Each disease is a unique case, for some no researched values exist, use proxies.

03

Apply to find the net effect

Multiply the risk value by the number of incidence and subtract from the baseline to get the change in incidences.

04

Scale yearly and US in total

130,000 women in US enter menopause before 45 every year. Scale the 100,000 net effect to fit.

Health Methodology

Disease	Risk Value	Annual Incidences	Net Effect	Scaled to 130,000
All Cause Mortality	1.12	750	90	117
Type 2 Diabetes	1.230	900	207	269.1
Cardiovascular Dis.	1.14	700	98	127.4
All Cause Dementia	1.15	300	45	58.5
Osteoporosis & Frac.	1.04	200	8	10.4
Breast Cancer	0.833	55	-9.185	-11.9405
Endometrial Cancer	0.962	13	-0.494	-0.6422
Ovarian Cancer	0.734	4	-1.964	-2.5532

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Economic Methodology

01

Find the annual direct cost per incidence

Every disease is a unique case. Search for closest, most reasonable proxies.

02

Find the annual indirect cost per incidence

Every disease is a unique case. Search for closest, most reasonable proxies.

03

Apply to 130,000 to find yearly costs

This cost can be treated as yearly addressable market.

04

Scale to 5 years and US in total

10% of women undergo menopause before 45, this is 17.1 million women. This can be seen as a total addressable market in US.

Economic Methodology

Disease	Annual Direct Cost	Annual Indirect Cost	Yearly 130,000	Present Value, 3%
All Cause Mortality	$80,000	$553,800	$74,154,600	$339,605,821
Type 2 Diabetes	$12,022	$13,630	$6,902,953	$31,613,454
Cardiovascular Dis.	$19,145	$36,500	$7,089,173	$32,466,285
All Cause Dementia	$53,502	$41,607	$5,563,876	$25,480,885
Osteoporosis & Frac.	$50,508	$5,848	$586,102	$2,684,173
Breast Cancer	$23,480	$9,156	-$389,690	-$1,784,664
Endometrial Cancer	$27,280	$12,500	-$25,546	-$116,996
Ovarian Cancer	$44,880	$10,000	-$75,910	-$347,645

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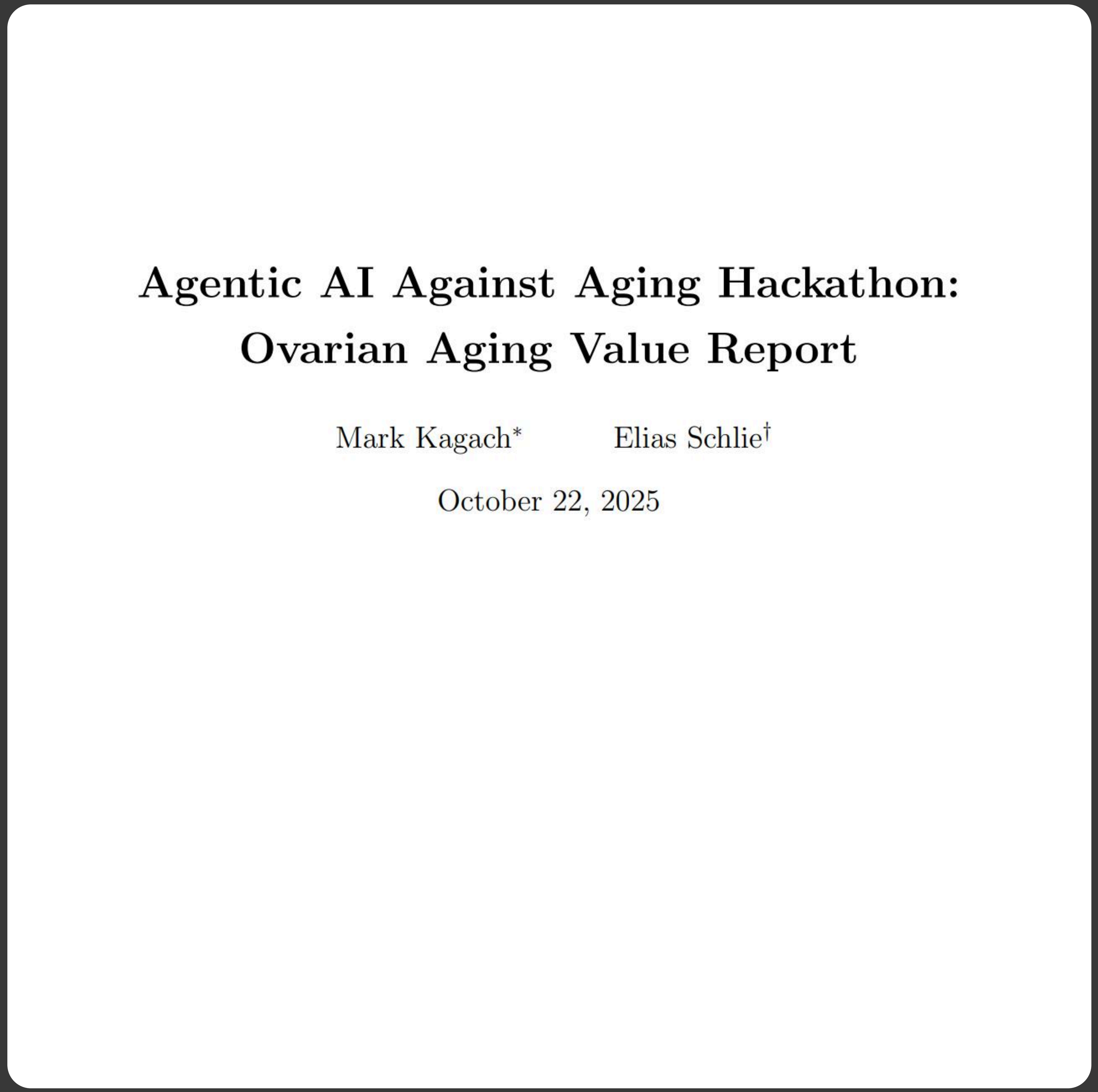
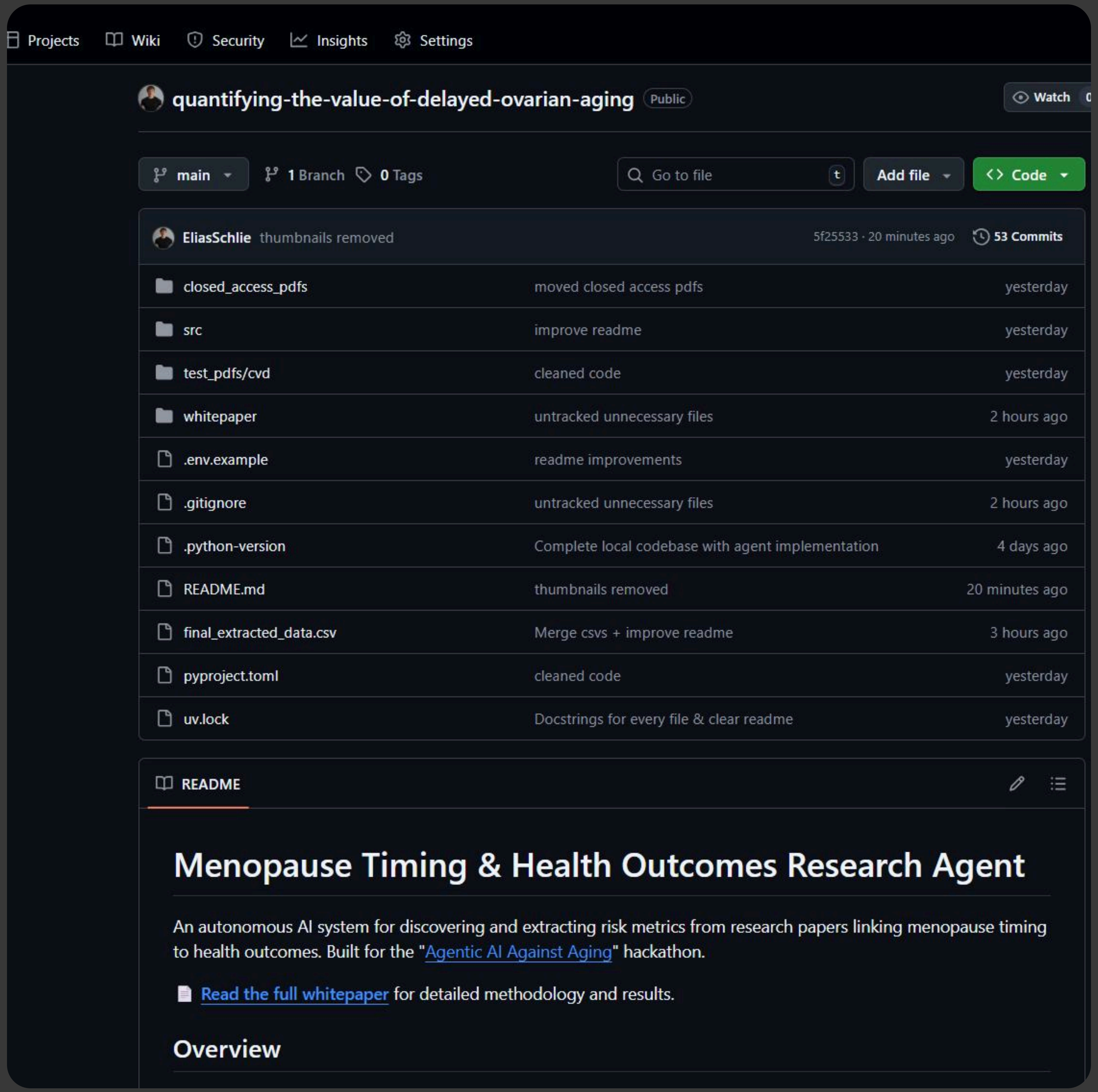

Agentic AI Against Aging Hackathon:
Ovarian Aging Value Report

Mark Kagach* Elias Schlie†

October 22, 2025

Report ↗



Code Repo ↗

MIT License*

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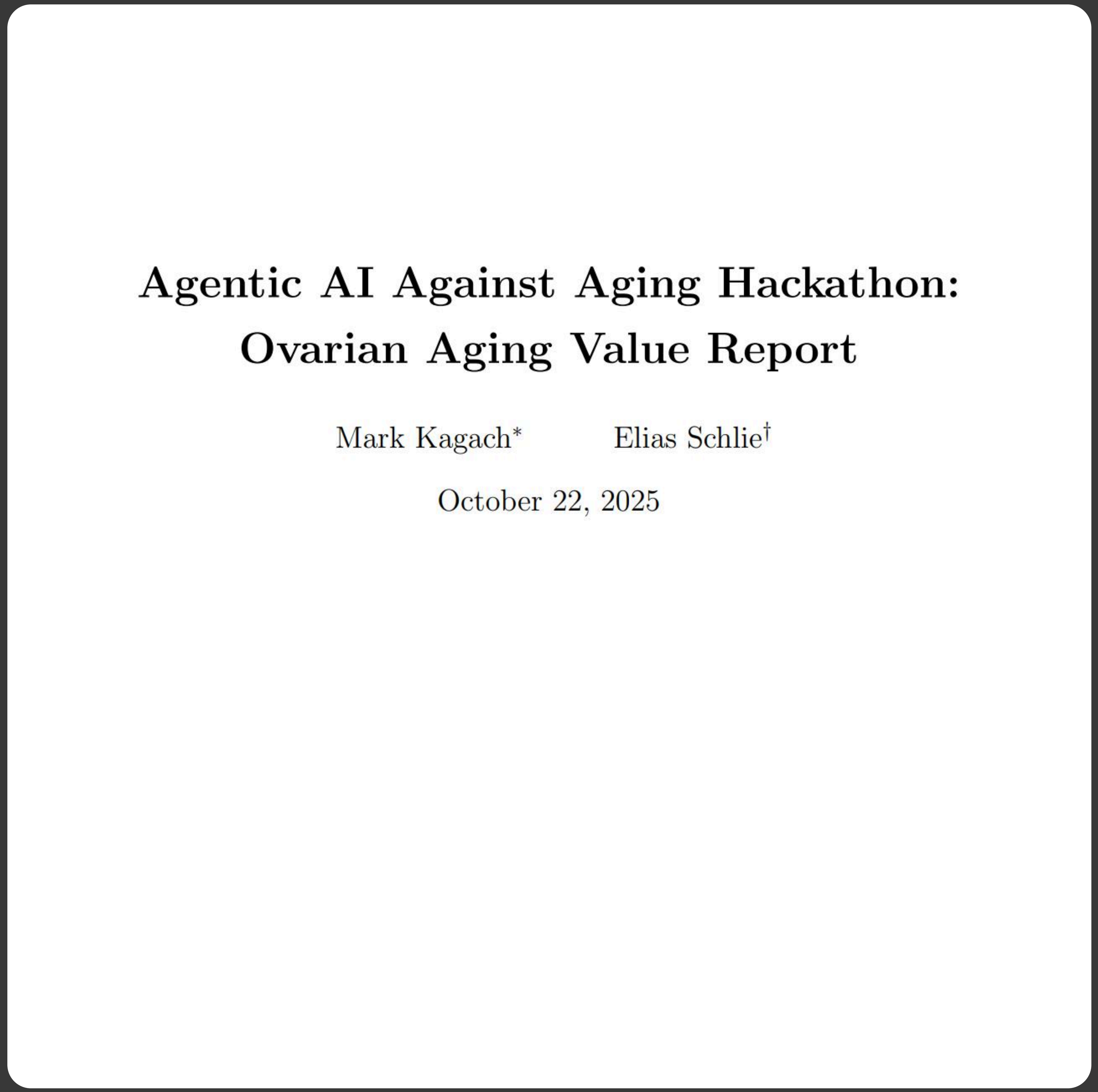

Agentic AI Against Aging Hackathon: Ovarian Aging Value Report

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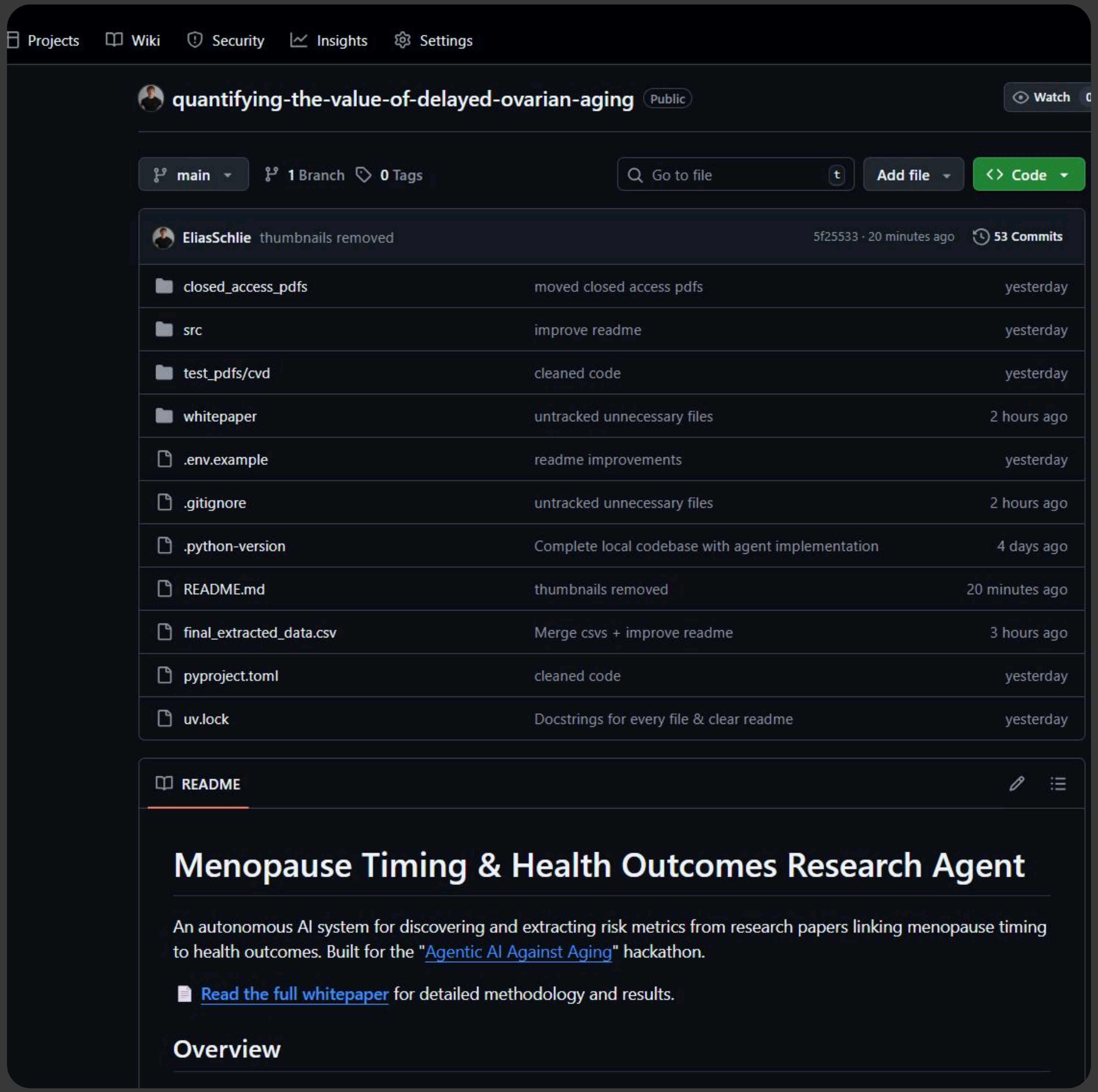
October 22, 2025

Report ↗



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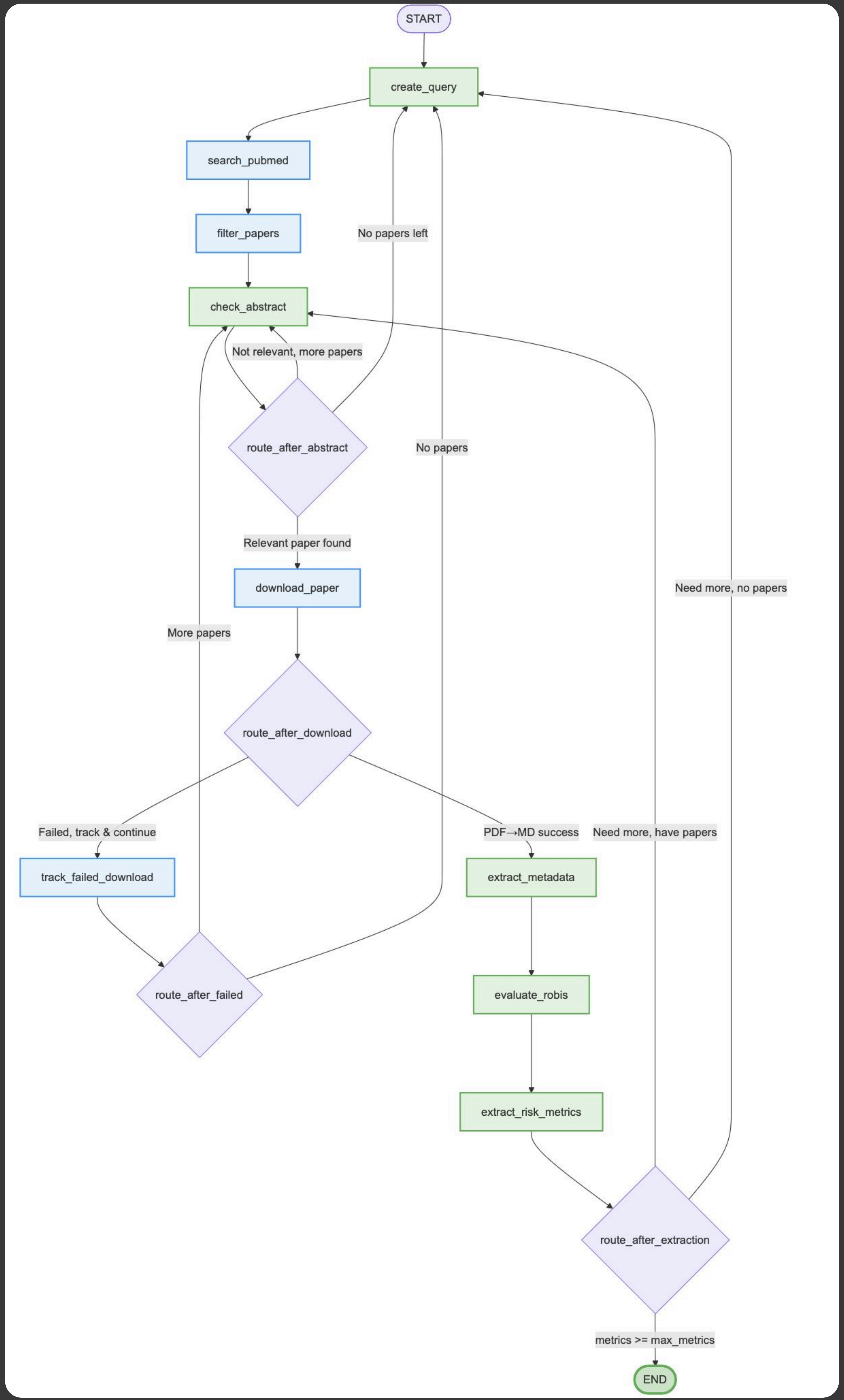

Agent Architecture

1. Query creation (AI)
2. PubMed API search
3. Checking abstract for relevance (AI)
4. Downloading relevant papers
5. Extracting paper metadata (AI)
6. Evaluating the paper based on ROBIS criteria (AI)
7. Extracting risk metrics (AI)